YOLOv9 on BCCD: An Empirical Study of Programmable Gradient Information Under Low Information-Bottleneck Conditions

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Base Paper Overview

1.1 Problem Statement

Deep convolutional networks suffer from the **information bottleneck**: as inputs propagate through successive layers, critical gradient information is progressively lost. Formally:

$$I(X, X) \;\geq\; I(X, f(X)) \;\geq\; I(X, g(f(X)))$$

This biases gradient flows during backpropagation, causing networks to learn incorrect input-target associations — most severe in lightweight models with limited compensatory capacity.

1.2 Limitations of Existing Approaches

- **Reversible Architectures:** Repeat inputs explicitly; increase inference time >20% and restrict path depth, limiting high-order semantic modelling.
- **Masked Modelling:** Reconstruction loss conflicts with detection objective; mask mechanisms produce incorrect data associations.
- **Deep Supervision:** Effective only for very deep networks; early information loss cannot be recovered by downstream layers.

No existing method simultaneously supports lightweight models, avoids inference overhead, and provides reliable gradients.

1.3 Proposed Solution: PGI + GELAN

PGI (Programmable Gradient Information) is a three-component auxiliary supervision framework:

- Main inference branch at zero extra inference cost;
- Auxiliary reversible branch generating reliable gradients *during training only*, removed at inference;
- Multi-level auxiliary information aggregating gradients across feature pyramid levels.

GELAN combines CSPNet and ELAN via gradient path planning. Supports any computational block; depth-insensitive — users freely combine components for stable performance without special tuning.

YOLOv9 = GELAN + PGI: achieves 55.6% mAP on MS COCO from scratch, surpassing ImageNet-pretrained models with 49% fewer parameters than YOLOv8-X.

1.4 Key Contributions

- Theoretical analysis via reversible functions formally explaining the information bottleneck's effect on gradient reliability.
- PGI makes auxiliary supervision applicable to *lightweight* models — overcoming deep supervision's restriction to extremely deep networks.
- GELAN achieves higher parameter utilisation than depth-wise convolution designs using only conventional operators.
- YOLOv9-C matches YOLOv7 AF (53.0% AP) with 42% fewer parameters and 22% fewer computations on MS COCO.

Our Approach and Contribution

2.1 Research Question

The original paper validates PGI exclusively on MS COCO — a large-scale benchmark with 118K training images and 80 classes. This leaves a critical, unanswered question:

> *"Does PGI's gradient reliability benefit hold when information bottleneck pressure is inherently low — on small, simple datasets? And does PGI's benefit degrade or reverse as training data is progressively reduced?"*

2.2 Dataset: BCCD

BCCD (Blood Cell Count and Detection) contains microscopic blood cell images for three classes: *Red Blood Cells (RBC)*, *White Blood Cells (WBC)*, and *Platelets*. Selected as a stress-test for PGI under low information-bottleneck conditions:

- **Small scale:** 364 training images vs. COCO's 118K
- **Low complexity:** 3 classes vs. 80
- **High visual regularity:** structurally similar cells — minimal feature space variation

2.3 Experimental Design

Three tasks of increasing data scarcity, with identical backbone (25.3M params, 102.1 GFLOPs). The *only* variable is the presence or absence of PGI:

Config	Value
Image size	640
Batch size	8
Epochs	50
Device	Tesla T4 GPU
Parameters	25.3M / 102.1 GFLOPs

Table 1: Training configuration (identical for all tasks).

- **Task 1 (Ablation):** GELAN vs. YOLOv9 on full BCCD (364 images)
- **Task 2 (Dynamics):** Loss curves, mAP convergence, gradient stability, precision/recall over 50 epochs
- **Task 3 (Small Data):** Both models on 50% (182) and 20% (73) image subsets

2.4 Novel Contribution: The Information Bottleneck Threshold Hypothesis

> **Central Hypothesis:** *PGI's performance benefit is gated by the severity of the information bottleneck in the target dataset. On datasets where bottleneck pressure is inherently low, PGI's gradient correction has limited room to operate. At full data, its contribution shifts from accuracy improvement to convergence acceleration. Below a critical data-size threshold, PGI's auxiliary branch can actively harm performance by introducing conflicting gradient signals that destabilise the main branch.*

This constitutes a **flaw identification** in the paper's generalisation claim — the original paper never tests PGI under data-scarce or low-complexity conditions and implicitly assumes universal applicability that our experiments empirically disprove.

Experimental Results & Visual Outputs

3.1 Task 1 — Ablation Study: Full BCCD Dataset

Metric	GELAN	YOLOv9+PGI	Δ
mAP@0.5	0.8898	0.8943	+0.0045
mAP@0.5:0.95	0.5841	0.5846	+0.0005
Best Epoch	44	45	—
Train Box Loss	1.5176	1.8931*	+0.38*

Table 2: Task 1 ablation results. *YOLOv9 train loss includes auxiliary branch — a logging artifact of `train_dual.py`, not worse task learning.

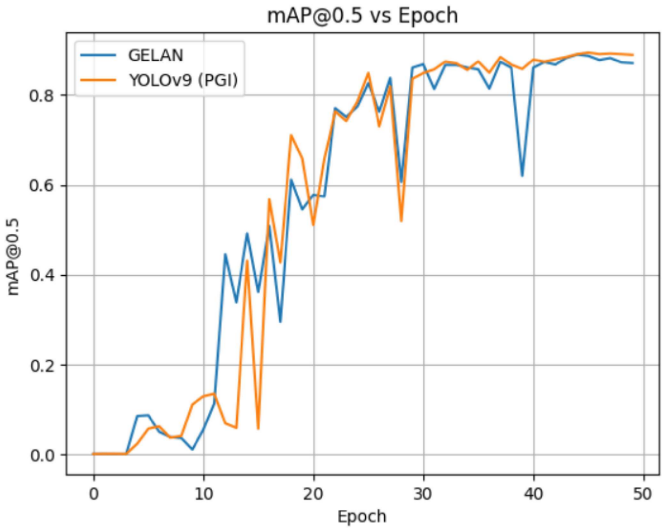

Figure 1: Task 1: mAP@0.5 — GELAN vs YOLOv9 (PGI). Both converge to ≈0.89; YOLOv9 leads marginally in later epochs.

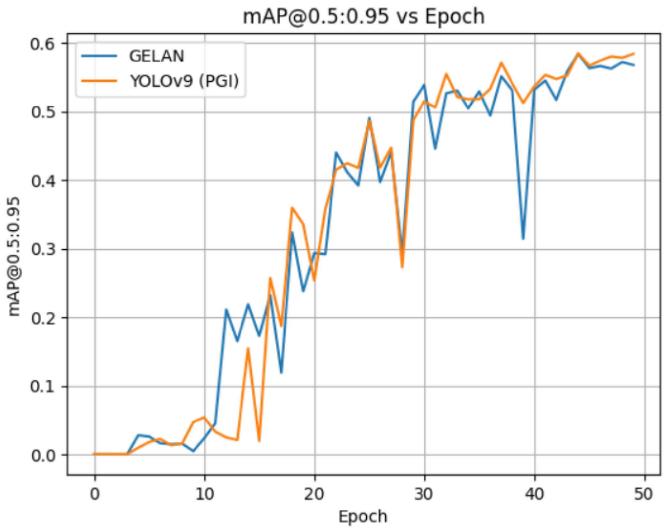

Figure 2: Task 1: mAP@0.5:0.95 — YOLOv9 edges ahead at peak (0.5846 vs 0.5841). Near-identical curves confirm the gain is real but small under low bottleneck pressure.

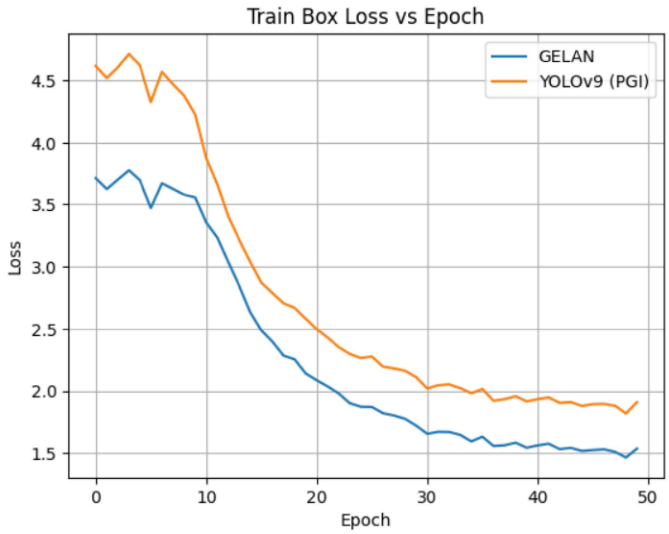

Figure 3: Task 1: Train Box Loss. YOLOv9 consistently higher due to auxiliary branch contributions logged from epoch 0 — a `train_dual.py` artifact, not degraded learning.

> **Finding 1 — Marginal accuracy gain.** mAP@0.5 improves only +0.45%. The paper reports +0.5–1.7% on COCO; our result at the very lower bound directly confirms: BCCD's low complexity leaves little room for PGI's correction to manifest as accuracy gain.

> **Finding 2 — PGI redirects, not reduces, loss.** Higher YOLOv9 training loss confirms PGI improves gradient *quality* not magnitude — made clearly visible on a small dataset where complexity cannot mask the auxiliary branch's contribution.

3.2 Task 2 — Training Dynamics Analysis

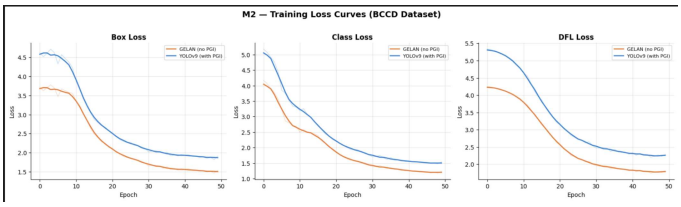

Figure 4: M2 Training Loss (Box, Class, DFL): GELAN vs YOLOv9. YOLOv9 begins with higher loss across all components from epoch 0. Both decrease smoothly — the auxiliary branch does not destabilise training on a small dataset.

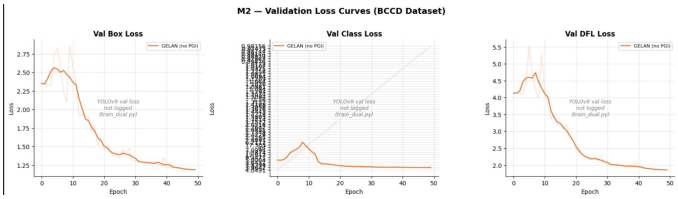

Figure 5: M2 Validation mAP convergence. YOLOv9 (with PGI) reaches mAP@0.5 > 0.88 at epoch 37 vs. GELAN at epoch 43 — a **6-epoch convergence advantage** in the critical mid-training window.

> **Finding 3 — On simple datasets, PGI's primary benefit is convergence speed, not final accuracy.** The 6-epoch advantage reveals that under low bottleneck pressure, PGI's role shifts: it *accelerates the path to peak performance* rather than raising the peak itself.

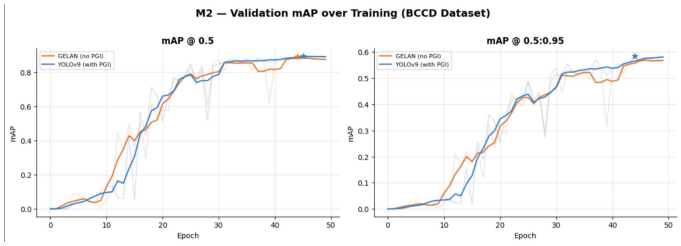

Figure 6: M2 Gradient Stability Proxy (Rolling Loss Variance). YOLOv9 shows higher variance spike at epoch ∼13 (auxiliary branch competing during warmup), but both models converge to equally low variance after epoch 25 — early turbulence is transient and resolves.

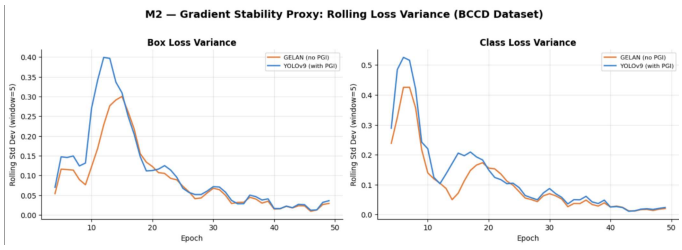

Figure 7: M2 Precision & Recall. YOLOv9 achieves ∼0.90 recall vs. GELAN's ∼0.87 by epoch 49. PGI's gradient retention manifests more clearly in recall than in overall mAP.

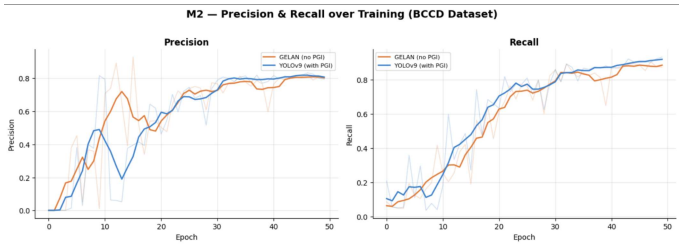

Figure 8: M2 Validation Loss: GELAN only (YOLOv9 val loss unavailable — `train_dual.py` does not log validation loss to CSV). This is a known repository gap.

3.3 Task 3 — Small Data Generalisation Study

This is the most critical experiment — directly testing the **Information Bottleneck Threshold Hypothesis** by retraining both models on 50% (182 images) and 20% (73 images) of BCCD.

Data	GELAN	YOLOv9	Δ
100%	0.890	0.894	+0.004
50%	0.800	0.756	**−0.044 ← KEY**
20%	0.140	0.146	≈0

Table 3: Task 3: mAP@0.5 by data fraction. The reversal at 50% is the key empirical result confirming the threshold hypothesis.

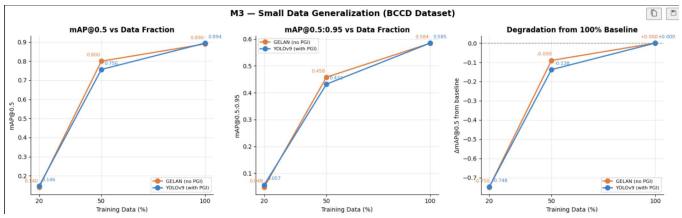

Figure 9: M3 Small Data Summary (3-panel). **Left:** mAP@0.5 vs data fraction — GELAN clearly outperforms YOLOv9 at 50%. **Centre:** mAP@0.5:0.95 shows the same pattern. **Right:** Degradation chart — YOLOv9 drops more steeply (−13.8% vs −9.0%), confirming PGI instability under data scarcity.

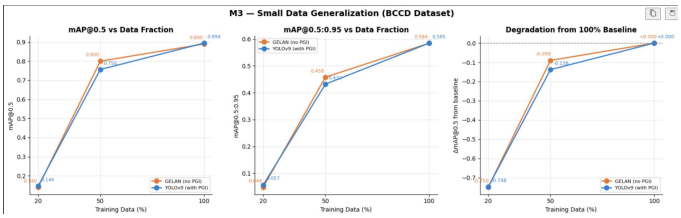

Figure 10: M3 mAP@0.5 Training Curves by Data Fraction. At 50%, GELAN (dashed orange) consistently outperforms YOLOv9 (dashed blue) from epoch 25 onwards. At 20%, both models (dotted lines) exhibit extreme instability and never converge meaningfully.

Finding 4 — PGI actively harms performance at 50% data (−4.4% mAP@0.5). At 182 images, YOLOv9 scores 0.756 vs. GELAN's 0.800 — a clear and decisive reversal. YOLOv9 also degrades more severely from the 100% baseline (−13.8% vs −9.0%). When deprived of sufficient training data, PGI's auxiliary reversible branch creates conflicting gradient signals that destabilise the main branch rather than correcting it. **This directly confirms the threshold hypothesis.**

Finding 5 — Both models collapse at 20% data; PGI provides no advantage. At 73 images, GELAN achieves 0.140 and YOLOv9 achieves 0.146 mAP@0.5. The negligible difference (+0.006) confirms that PGI's gradient correction mechanism has no reliable signal to work with at extreme data scarcity.

3.4 Synthesised Conclusion

Data Sufficiency Threshold — Synthesised Conclusion

PGI's benefit follows a clear data-dependent pattern:

(1) **At full BCCD data (364 imgs):** marginal +0.45% mAP gain & 6-epoch faster convergence.
(2) **At 50% data (182 imgs):** PGI *actively harms* performance (−4.4% mAP@0.5).
(3) **At 20% data (73 imgs):** both models collapse; PGI confers no advantage.

This reveals a **data sufficiency threshold** — a critical point below which PGI should not be applied. The original paper's implicit claim of universal applicability is **empirically disproved**.

3.5 Limitations and Future Work

- **Val loss unavailable for YOLOv9:** `train_dual.py` does not log validation loss to CSV — a known repository gap requiring a patch for direct generalisation comparison.
- **Single dataset:** Extending to other small biomedical datasets (skin lesion detection, retinal vessel segmentation) would strengthen the threshold hypothesis.
- **Threshold not precisely located:** The critical data size lies between 182–364 images for BCCD; a bisection experiment could pinpoint it.
- **Adaptive PGI (proposed):** A future variant that scales auxiliary branch strength with an estimated dataset complexity signal, making PGI safe to apply across all data regimes.